

Softmax Regression

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강의 내용

- 특징의 유형을 구분하여 설명할 수 있다.
- 소프트맥스(softmax) 함수에 대해 설명할 수 있다.
- 원-핫 인코딩(one-hot encoding)에 대해 설명할 수 있다.
- Cross Entropy에 대해 설명할 수 있다.

특징(feature)의 유형

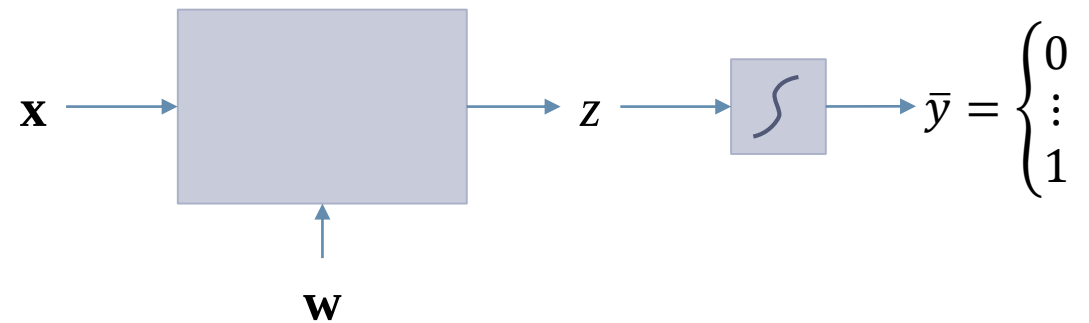
- 연속 수치형 ~ continuous numerical type
 - 꽃잎 길이(petal length), 몸무게(length of body) 등
- 순서 있는 범주형 ~ ordinal (discrete/ranked) type
 - 학년: 초등1~6/중등1~3/고등1~3
 - 서비스 만족도: Good, Neutral, Bad
- 순서 없는 범주형 ~ nominal (categorical) type (순서 무)
 - 정상/불량 (카드 사용)
 - virginica/setosa/versicolor (붓꽃 품종)
 - Korea/US/France/... (국가)
 - Black/Brown/Silver (머리카락 색상)

Logistic regression (1)

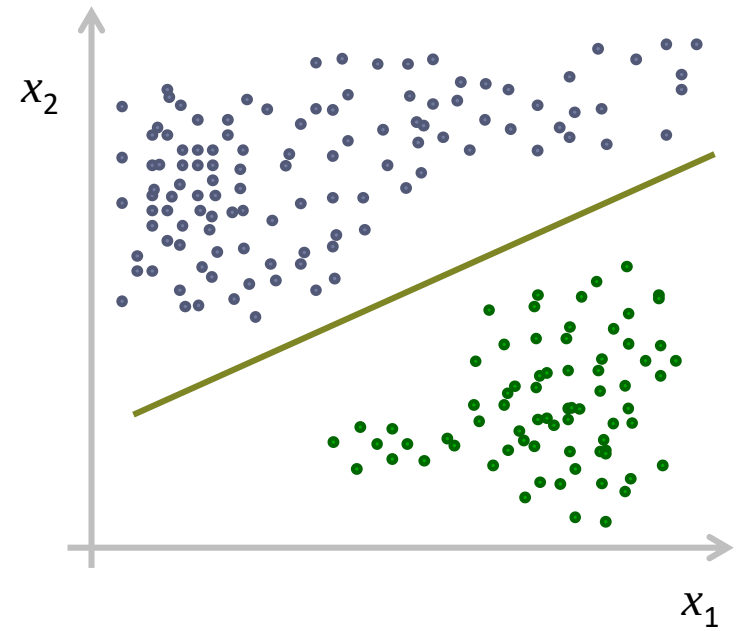
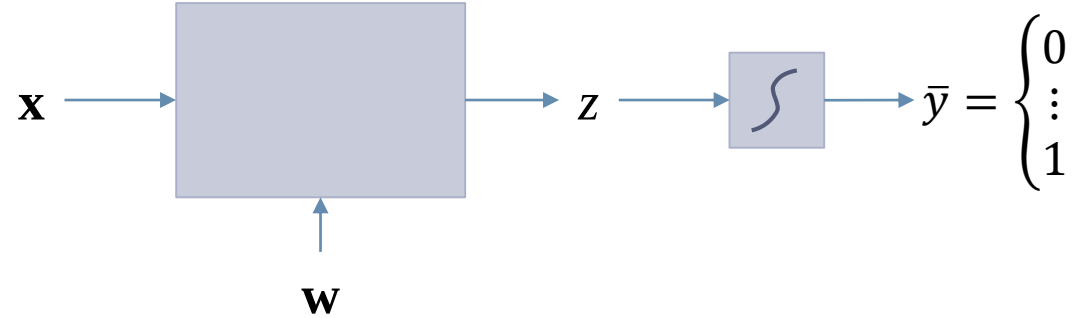
$$z = \mathbf{w}^T \mathbf{x}$$

$$H(\mathbf{x}) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}}}$$

$$J(\mathbf{w}) = \frac{1}{n} \sum_{k=1}^n -y^{(k)} \cdot \log(H(x^{(k)})) - (1 - y^{(k)}) \log(1 - H(x^{(k)}))$$

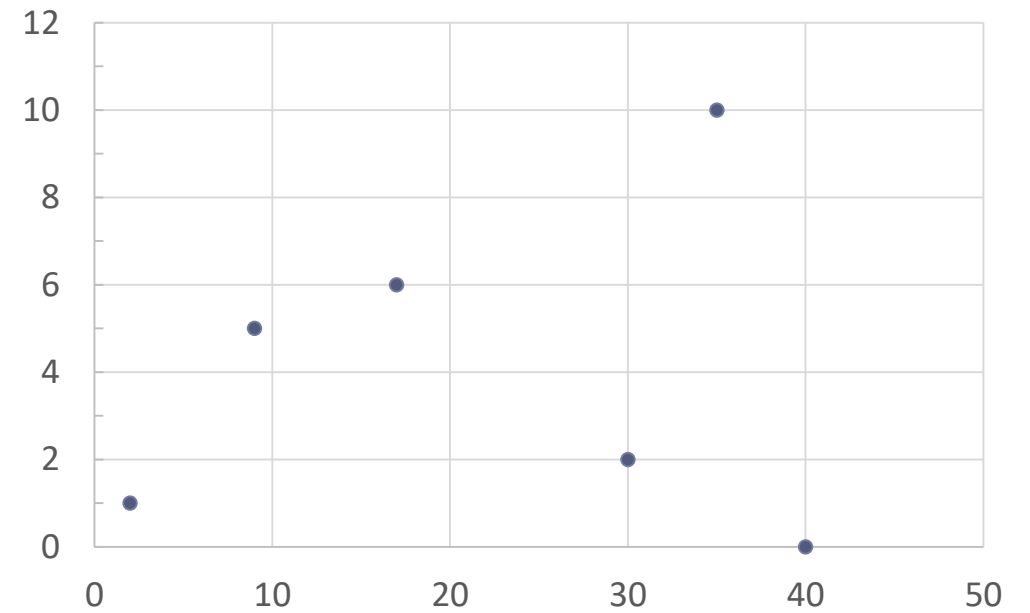


Logistic regression (2)

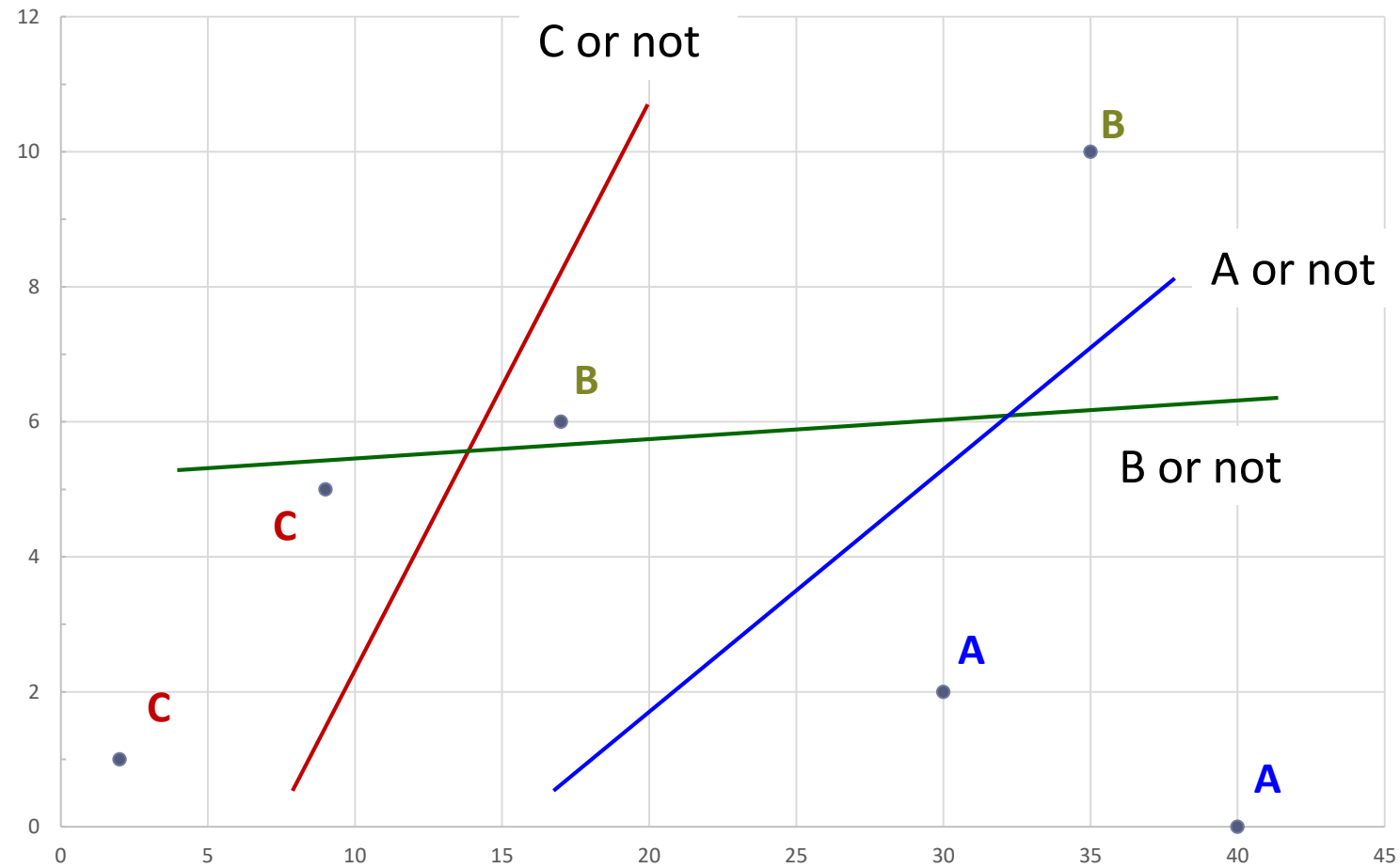


Multinomial classification (1)

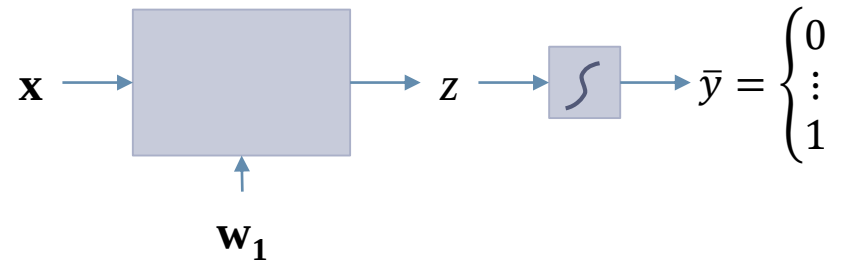
공부 시간(x_1)	결석일(x_2)	과목 등급(y)
9	5	C
30	2	A
17	6	B
2	1	C
40	0	A
35	10	B



Multinomial classification (2)

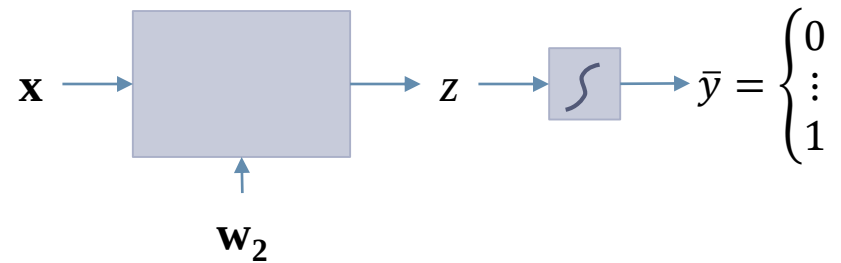


Multinomial classification (3)



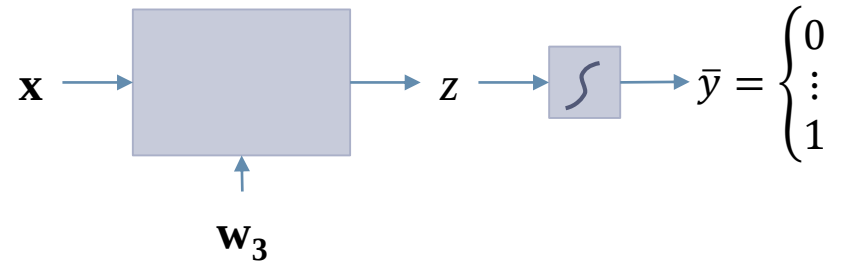
A

$$[w_{11} \quad w_{12}] \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = w_{11}x_1 + w_{12}x_2$$



B

$$[w_{21} \quad w_{22}] \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = w_{21}x_1 + w_{22}x_2$$



C

$$[w_{31} \quad w_{32}] \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = w_{31}x_1 + w_{32}x_2$$

Multinomial classification (4)

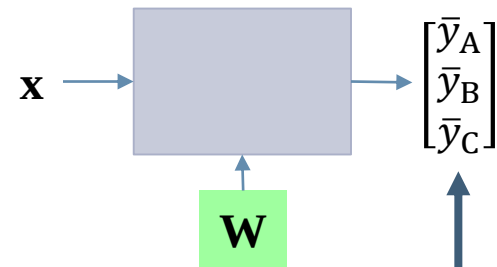
$$\begin{bmatrix} w_{A1} & w_{A2} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = w_{A1}x_1 + w_{A2}x_2$$

$$\begin{bmatrix} w_{B1} & w_{B2} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = w_{B1}x_1 + w_{B2}x_2$$

$$\begin{bmatrix} w_{C1} & w_{C2} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = w_{C1}x_1 + w_{C2}x_2$$

$$\begin{bmatrix} w_{A1} & w_{A2} \\ w_{B1} & w_{B2} \\ w_{C1} & w_{C2} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} w_{A1}x_1 + w_{A2}x_2 \\ w_{B1}x_1 + w_{B2}x_2 \\ w_{C1}x_1 + w_{C2}x_2 \end{bmatrix} = \begin{bmatrix} \bar{y}_A \\ \bar{y}_B \\ \bar{y}_C \end{bmatrix}$$

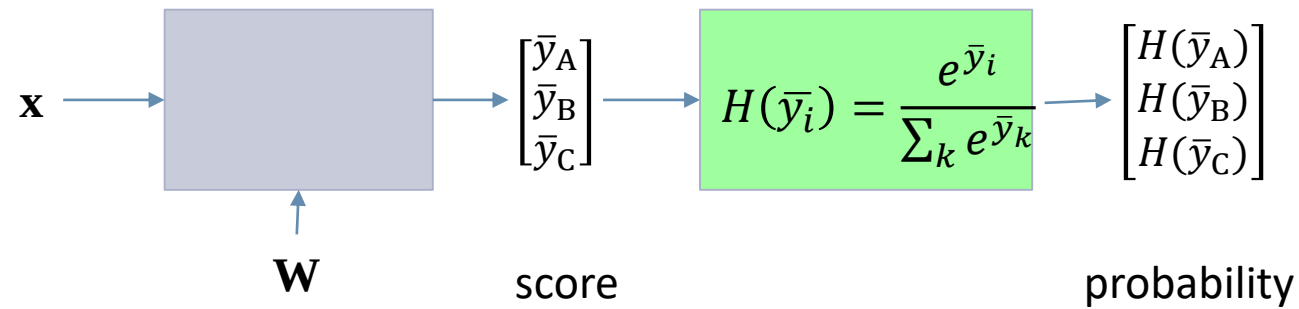
Sigmoid 함수는?



각 출력이 0~1이고
모두 합쳐서 1이 되도록

Softmax

$$\begin{bmatrix} w_{A1} & w_{A2} \\ w_{B1} & w_{B1} \\ w_{C1} & w_{C1} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} w_{A1}x_1 + w_{A2}x_2 \\ w_{B1}x_1 + w_{B2}x_2 \\ w_{C1}x_1 + w_{C2}x_2 \end{bmatrix} = \begin{bmatrix} \bar{y}_A \\ \bar{y}_B \\ \bar{y}_C \end{bmatrix}$$



One-hot encoding

$$\bar{\mathbf{y}} = \begin{bmatrix} 7.0 \\ 2.0 \\ 0.5 \end{bmatrix} \rightarrow H(\bar{\mathbf{y}}) = \begin{bmatrix} 0.74 \\ 0.21 \\ 0.05 \end{bmatrix} \rightarrow R(\bar{\mathbf{y}}) = \begin{bmatrix} 1.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

속성

클래스

	정보 1	정보 2	정보 3	정보 4	품종
1번째 아이리스	5.1	3.5	4.0	0.2	Iris-setosa
2번째 아이리스	4.9	3.0	1.4	0.2	Iris-setosa
3번째 아이리스	4.7	3.2	1.3	0.3	Iris-setosa
...	...	...	...	...	...
150번째 아이리스	5.9	3.0	5.1	1.8	Iris-virginica

샘플

1 [0., 1., 0.]

1 [0., 1., 0.]

1 [0., 1., 0.]

0 [1., 0., 0.]

Softmax Hypothesis

$$S(\mathbf{z}) = \frac{e^{\mathbf{z}}}{\sum e^{\mathbf{z}}}$$

$$\mathbf{z} = W\mathbf{x}$$

$$H(\mathbf{x}) = \frac{e^{W\mathbf{x}}}{\sum e^{W\mathbf{x}}}$$

$R(\mathbf{x})$: one-hot encoding of $H(\mathbf{x})$

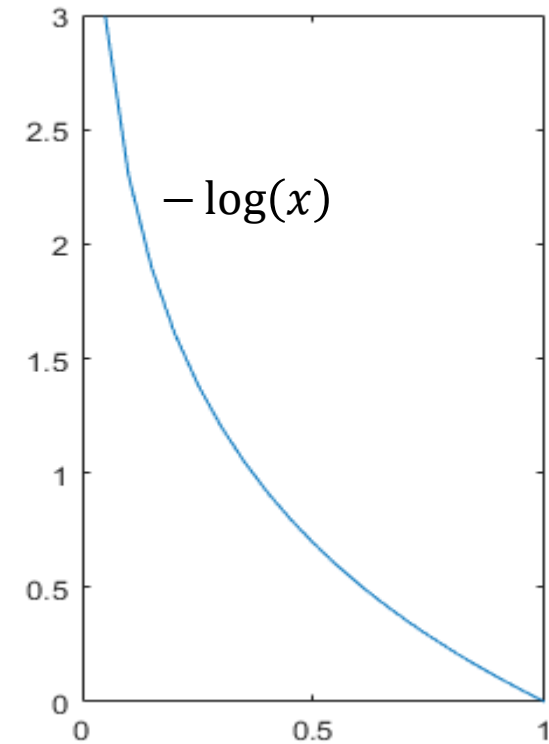
Loss function

$$loss(\bar{\mathbf{y}}) = - \sum_i y_i \cdot \log(\bar{y}_i)$$

Cross entropy

Loss entropy (1)

$$\begin{aligned} loss(\bar{\mathbf{y}}) &= - \sum_i y_i \cdot \log(\bar{y}_i) \\ &= \sum_i y_i \cdot (-\log(\bar{y}_i)) \end{aligned}$$



Cross entropy (2)

[정답]

$$\mathbf{y} = \begin{bmatrix} 1.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

[예측 경우1]

$$\bar{\mathbf{y}} = \begin{bmatrix} 0.9 \\ 0.0 \\ 0.1 \end{bmatrix}$$

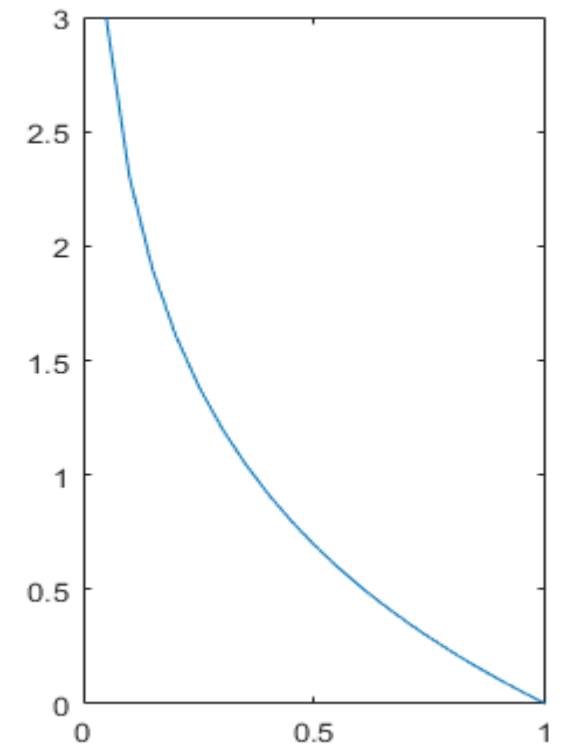
cost는 0 부근이 됨

[예측 경우2]

$$\bar{\mathbf{y}} = \begin{bmatrix} 0.0 \\ 0.9 \\ 0.1 \end{bmatrix}$$

cost는 ∞ 가 됨

$$\begin{aligned} \text{loss}(\bar{\mathbf{y}}) &= - \sum_i y_i \cdot \log(\bar{y}_i) \\ &= \sum_i y_i \cdot (-\log(\bar{y}_i)) \end{aligned}$$

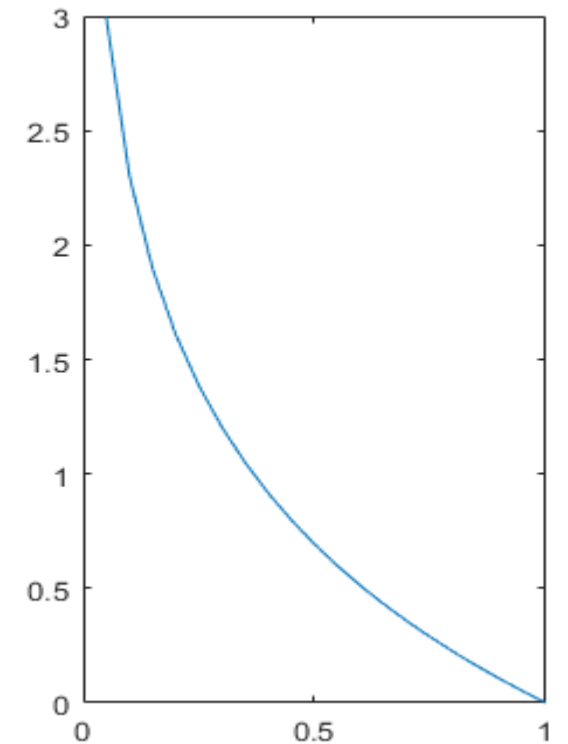


Logistic cost vs. Cross entropy

$$loss(\bar{y}) = -y \cdot \log(\bar{y}) - (1 - y)\log(1 - \bar{y})$$

$$loss(\bar{\mathbf{y}}) = -\sum_i y_i \cdot \log(\bar{y}_i)$$

형태는 다르지만 의미는 동일함



Cost function

$$loss(\bar{\mathbf{y}}) = - \sum_i y_i \cdot \log(\bar{y}_i)$$

$$J(\mathbf{w}) = \frac{1}{n} \sum_{k=1}^n loss(\bar{\mathbf{y}}) = \frac{1}{n} \sum_{k=1}^n \left(- \sum_i y_i^{(k)} \cdot \log(\bar{y}_i^{(k)}) \right)$$

Minimize cost

$$J(\mathbf{w}) = \frac{1}{n} \sum_{k=1}^n \text{loss}(\bar{\mathbf{y}}) = \frac{1}{n} \sum_{k=1}^n \left(- \sum_i y_i^{(k)} \cdot \log(\bar{y}_i^{(k)}) \right)$$

Gradient descent:

$$\mathbf{w}^{(t+1)} := \mathbf{w}^{(t)} - \alpha \frac{\partial}{\partial w_{ki}} J(\mathbf{w})$$

for $k = 0, 1, 2, \dots, n$

for $i = 0, 1, 2, \dots, m$

코딩으로 확인하는 소프트맥스 회귀

- 016_01_softmax_regression_sklearn.ipynb
- 016_02_softmax_regression_TF.ipynb

참고자료

- 모두를 위한 머신러닝/딥러닝 (홍콩과기대 김성훈교수)

- <http://hunkim.github.io/ml/>

- Stanford Machine Learning

- <http://www.holehouse.org/mlclass/>

- Deep Learning by Google

- <https://www.udacity.com/course/deep-learning--ud730>

- Machine Learning (한성대 지준교수)

- <http://jun.hansung.ac.kr/ml>

다음 강의 내용

- 분류 평가 지표